CUNY Queens College Department of Computer Science
CSCI 212 Lab - Object Oriented Programming in Java

Project 2

Advanced OOP with Pokémon

Due Date: April 12, 2026 11:59pm

Objective: In this project, you will design a program that reads data from a text file, utilizes Inheritance and Interfaces to create distinct Pokémon objects, uses Lambda expressions for sorting, and displays the final roster in a Java Swing Graphical User Interface (GUI). This project builds upon the foundational Java concepts covered in Project 1.

Project Requirements

You must create the following Java files to ensure your project is properly modularized.

Class 1: Move.java Interface

Create an interface named Move. This interface will ensure that all playable Pokémon have a defined set of attacks.
Methods:
Public String[] getMoveList()
It should contain a single method signature.

Class 2: Pokemon.java Superclass

Create a class to serve as the blueprint for all Pokémon.
Instance variables:

- Private name: String
- Private type: String
- Private height: double
- Private weight: double
- Private description: String

Constructor: Create a constructor that accepts all five of these parameters to initialize a new Pokémon object.
If there is a type mismatch, then throw an IllegalArgumentException.
Methods: Include standard Getters and Setters for all variables.
Include a toString() method to cleanly format all the Pokémon's information into one nice String

Class 3,4,5,6: The Elemental Subclasses

Create four distinct subclasses: ElectricPokemon.java, FirePokemon.java, WaterPokemon.java, and GrassPokemon.java.
Each class must extend the Pokemon superclass and implement the Move interface.
Constructors: Each subclass constructor should accept the necessary parameters and pass them to the parent class using the super() keyword.
Interface Implementation: Each class must implement the getMoveList() method to return a String array containing four unique elemental moves.
Here are some ideas:
Electric: "Thunderbolt", "Thunder", "Spark", "Volt Tackle"
Fire: Ember, Flame Wheel, Flamethrower, Fire Blast
Water: Water Gun, Water Pulse, Surf, Hydro Pump
Grass: Vine Whip, Razor Leaf, Giga Drain, Solar Beam

Class 7: Main.java Driver and File I/O

This class will manage the execution of your program.
File Reading: Your program must read from a provided text file named pokemon.txt.
Parsing Data: Read the file line-by-line. The data is Comma-Separated (CSV). Use the String.split(",") method to break each line into its respective components.
Object Instantiation: Based on the "Type" read from the file, instantiate the correct subclass (e.g., if the type is "Electric", create an ElectricPokemon object) and add it to an ArrayList<Pokemon>.
Before displaying the Pokémon, you must sort your ArrayList.
Use the Collections.sort() or List.sort() method combined with a Lambda expression to sort the list of Pokémon alphabetically by their name.
Instantiate a NEW GUI OBJECT called pokemonDisplay. You will pass this sorted array list into your GUI constructor for the display.

Class 8: The Display GUI

Create a simple, non-interactive visual display using Java Swing.
Inside the GUI class, create a constructor that will take in the Array List passed in through main. It will be saved to a private attribute called infoToDisplay.
Create a JFrame with a GridLayout (e.g., 2x2 to fit four Pokémon).
Loop through your sorted ArrayList that was passed into the GUI. For each Pokémon, create a JPanel.
Set the background color of the JPanel based on the Pokémon's elemental type (e.g., Color.YELLOW for Electric, Color.RED for Fire, etc.).
Add JLabel components to the panel to display the Pokémon's name, stats, description, (coming from the passed in sorted array list) and their four moves (retrieved using the getMoveList() method).
Add each panel to the main frame and make it visible.
You can be as creative as you want with the formatting, just make sure you include all the data! :)

Sample pokemon.txt Data

Please copy and paste this into a file named pokemon.txt in your project directory to use for reading into your code with the Scanner:

```
Pikachu,Electric,1.4,13.2,The Electric Mouse Pokemon
Charmander,Fire,2.0,18.7,The Lizard Pokemon
Squirtle,Water,1.8,19.8,The Tiny Turtle Pokemon
Bulbasaur,Grass,2.3,15.2,The Seed Pokemon
```

Submission Guidelines

Submit your EIGHT .java files (Main, Pokemon, Move, and the four subclasses).
Submit a screenshot of your GUI output with all FOUR panels and all of the information from the text file displayed as the instructions state.